

Shenzhen K&D Technology Co. Ltd

☒ Preliminary Specification

☐ Approval Specification

SPECIFICATION FOR LCD MODULE

Customer : _____

Product Model: _____ KD070D82-39TI-A009

Sample code: _____

Designed by	Checked by	Approved by
Xiaoling Wei	Xiao Yang	Xiyang Zhang

Final Approval by Customer

☐ LCM Machinery OK Checked By ☐ LCM Display OK Checked By	☐ LCM OK ☐ NG, Problem survey: Approved By
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※ The specification of "TBD" should refer to the measured value of sample . If there is difference between the design specification and measured value, we naturally shall negotiate and agree to solution with customer.

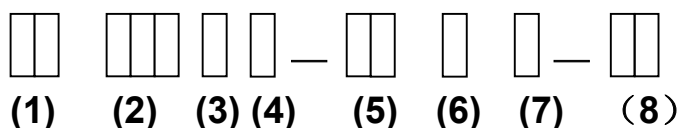
Revision History

[illegible]

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1. Numbering System



No	Definition	Specifications
(1)	TFT LCM Productor No.	KD ---- Kingdisplay technology Co.,Ltd
(2)	Display monitor opposite angle line size	Unit :mm (takes three integers)
(3)	Productor Types	D ---- PMP / Tablet PC G ----GPS M ----MP P ----Mobile-Phone N ----Net Book/MID ○ ○ ○
(4)	Productor Development Series No.	By two figures characters expression from 1 to 99
(5)	Interface PIN Number	By two figures characters expression from 1 to 99
(6)	With Touch Panel Or Not	T----With T/P ; N----Without T/P
(7)	LCD Type	A----AUO ; M----CMI ; C----CPT; B----BOE; L----LG; W----Wintek; H----HSD; S----Century T----Tianma ; Y----Hydis ; I----INNOLUX ; U---Samsung; V---- IVO; P---- Panasonic
(8)	Productor Development edition No.	By The English letters : A1~ Z999

2. Scope

This specification applies to the TFT LCD module which is designed and manufactured by LCM Factory of Shenzhen K&D Technology Co. Ltd.

3. Normative Reference

GB/T4619-1996 《Liquid Crystal Display Test Method》

GB/T2424 《Basic environmental Testing Procedures for Electric and Electronic Products.》

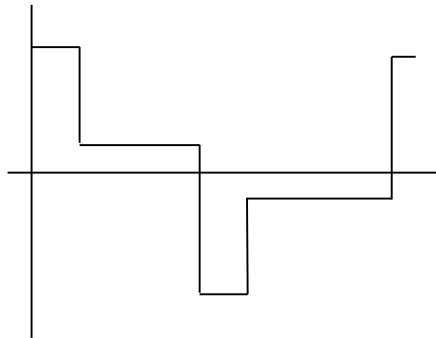
GB/T2423 《Basic Testing Procedures for Electric and Electronic Products》

IEC61747-1 《SIXTH PARTGB2828`2829-87 《National Standard of PRC》

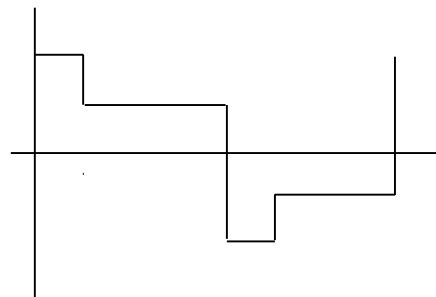
4. Definitions

4.1 Definitions of Vop

The definitions of threshold voltage Vth1, Vth2 the following typical waveforms are applied on liquid crystal by the method of equalized voltage for each duty and bias.



【 selected waveform 】



【 non-selected waveform 】

① Vth1: The voltage which the brightness of segment indicates 50% of saturated value on the conditions of selected waveform

($f_r=80\text{Hz}$, $\Phi=10^\circ$ $\theta=270^\circ$ at 25°C)

② Vth2: The voltage which the brightness of segment indicates 50% of saturated value on the conditions of non-selected waveform

($f_r=80\text{Hz}$, $\Phi=10^\circ$ $\theta=270^\circ$ at 25°C)

③ Vop: $(V_{th1}(50\%)+V_{th2}(50\%))/2$ ($f_r=80\text{Hz}$, $\Phi=10^\circ$ $\theta=270^\circ$ at 25°C)

4.2 Definition of Response Time Tr, Td

①Tr: The time required which the brightness of segment becomes 10% from 100% when waveform is switched to selected one from non-selected one. ($f_r=80\text{Hz}$, $\Phi=10^\circ$ $\theta=270^\circ$ at 25°C)

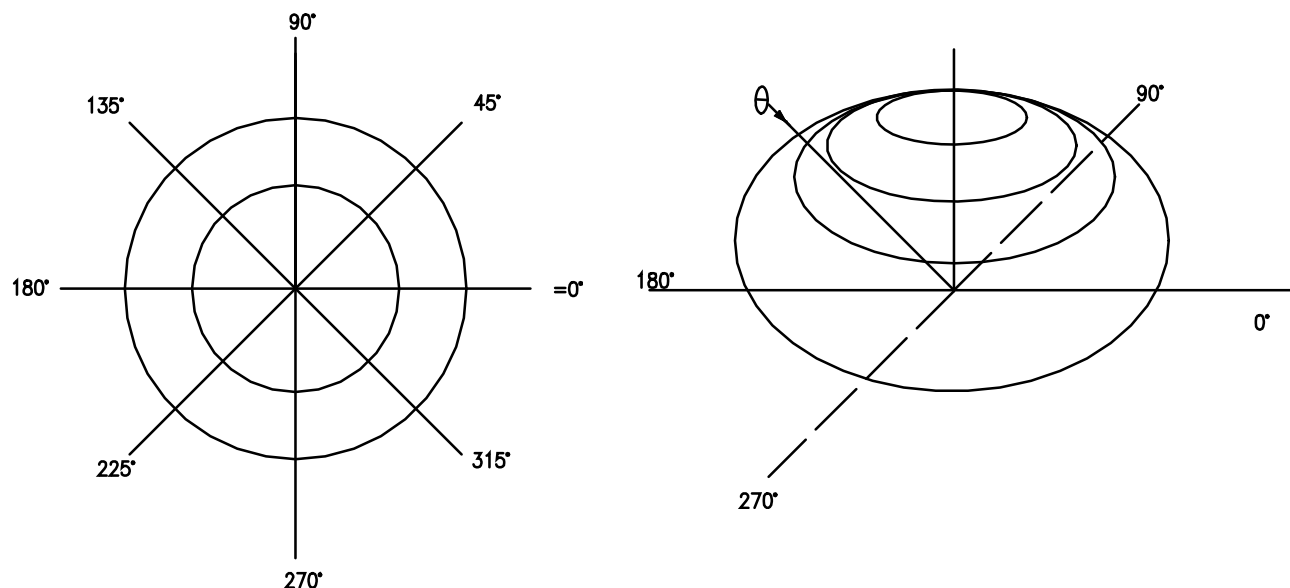
②Td: The time required which the brightness of segment becomes 90% from 10% when waveform is switched to selected one from selected one. ($f_r=80\text{Hz}$, $\Phi=10^\circ$ $\theta=270^\circ$ at 25°C)

4.3 Definition of Contrast Ratio Cr

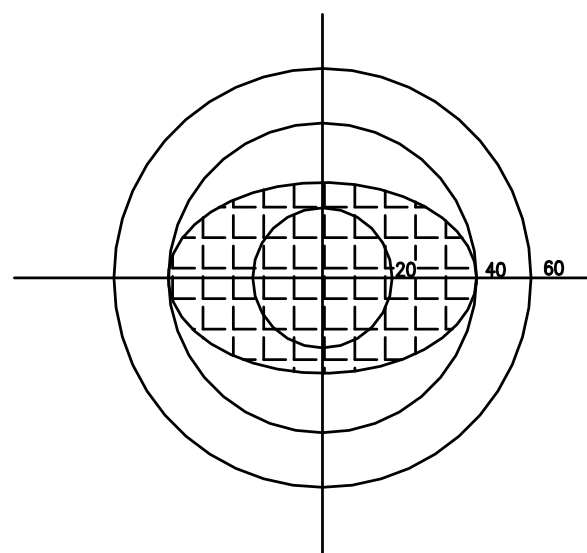
$$Cr=A/B$$

- ① A: Segments brightness in case of non-selected waveform
- ② B: Segments brightness in case of selected waveform

4.4 Definition of Angle and Viewing Range

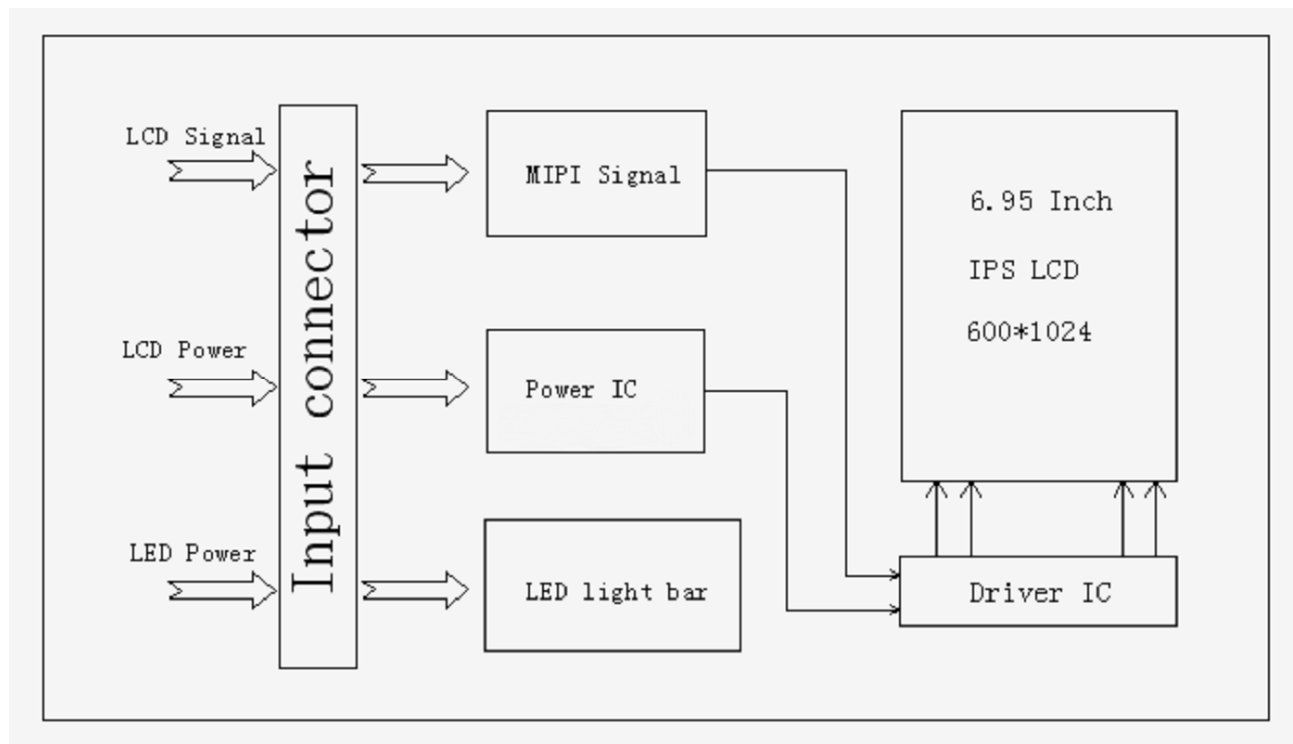


Angular Graph: Constrast Ratio



Such as:
Viewing Angle Range:
80($Cr>2$) Horizontal
70($Cr>2$) Vertical

5. Block Diagram



6. Technology Specifications

6.1 Features

This single-display module is suitable for use in MID products. The LCD adopts one backlight with High brightness 18-lamps white LED. Construction: 6.95 " a-Si color TFT-LCD ,With INX Cell, White LED backlight,CG and FPC .

6.2 General Specifications

No.	Item	Specification
1	LCD size	6.95 inch
2	Resolution	600 (RGB)X1024
3	Display mode	Normally black
4	Pixel pitch	0.1488(H)X0.1488(V) mm
5	Active area	89.28 (H)X152.37(V) mm
6	Module size(LCM+CG)	105.42(H)X171.05(V)X3.72(T) mm
7	Pixel arrangement	RGB-stripe
8	Surface Treatment	HC
9	Interface	MIPI
10	Backlight power consumption	1.14W(Typ)
11	LCM power consumption	0.18w (typ.)@RGB pattern
12	Weight	111g+/-10g

6.3 Interface Pin Connection

Pin No.	Symbol	I/O	Function
1	VLED+	P	Anode for light bar
2	VLED+	P	Anode for light bar
3	NC		NC
4	FB1	P	Cathode for light bar
5	FB2	P	Cathode for light bar
6	FB3	P	Cathode for light bar
7	NC		nc
8	TP_RST	I	Dedicated active low external reset pin 1.8V
9	TP_ATTN	I/O	Spare GPIO (may be used for ATTN)
10	TP_SDA	I/O	I2C data (SDA) or SPI data input (MOSI)
11	TP_SCL	I/O	I2C clock (SCL) or SPI clock (SCLK)
12	GND	P	Ground
13	NC		NC
14	VDDIN_TP	I	3.1V input for TP
15	GND	P	Ground
16	RESET	I	Device reset signal for LCM
17	LEDPWM	O	PWM control signal for LED driver (CABC). If no used, please keep it floating
18	NC		NC
19	ID1 (GND)		Ground
20	ID2(1.8V)		10K RES to 1.8V
21	GND		Ground
22	VDD3V3	P	3.3V power input for LCM
23	VDD3V3	P	3.3V power input for LCM
24	GND	P	Ground
25	D0N	I	MIPI differential data0 input (Negative)
26	D0P	I	MIPI differential data0 input (Positive)
27	GND	P	Ground
28	D1N	I	MIPI differential data1 input (Negative)
29	D1P	I	MIPI differential data1 input (Positive)
30	GND	P	Ground
31	CLKN	I	MIPI differential clock input (Negative)
32	CLKP	I	MIPI differential clock input (Positive)
33	GND	P	Ground
34	D2N	I	MIPI differential data2 input (Negative)
35	D2P	I	MIPI differential data2 input (Positive)
36	GND	P	Ground
37	D3N		NC
38	D3P		NC
39	GND	P	Ground

6.4 Absolute Max. Rating

Item	Symbol	Values		Unit
		Min.	Max.	
Power Voltage	VDD	-0.5	5.0	V
Backlight forward current	I _{LED}	0	30	mA(For each LED)
Input Signal Voltage	V _I	-0.3	VDD	V
Operation Temperature	T _{OP}	-20	60	°C
Storage Temperature	T _{ST}	-30	70	°C

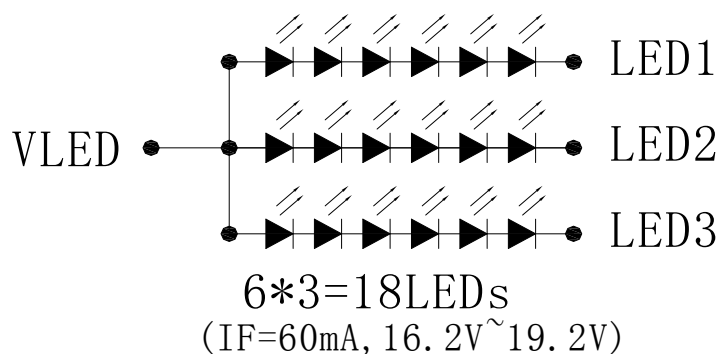
6.5 Typical Operation Conditions

Item	Symbol	Values			Unit
		Min.	Typ.	Max.	
Power Voltage	VDD	3.2	3.3	3.4	V
Current Consumption	I _{VDD}	-	55	-	mA Max(Red)

6.6 LED Back Light Specification (18 White Chips)

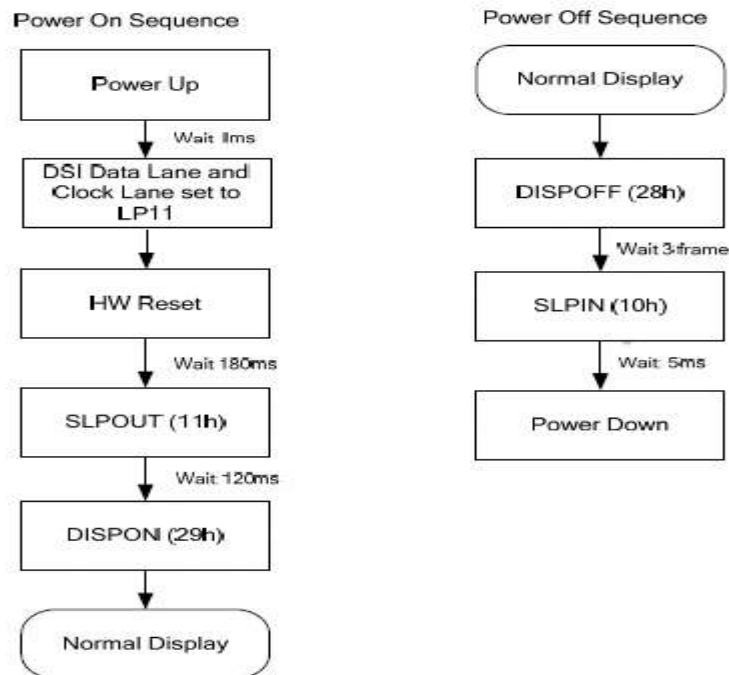
Item	Symbol	Condition	Min	Typ	Max	Unit
Forward Voltage	V _f	I _f =60mA	16.2	-	19.2	V
Uniformity (with L/G)	Δ B _p	I _f =60mA	75	80	-	%
Luminance for LCM+CG	/	I _f =60mA	300	350	-	cd/m ²

Note:LED Circuit



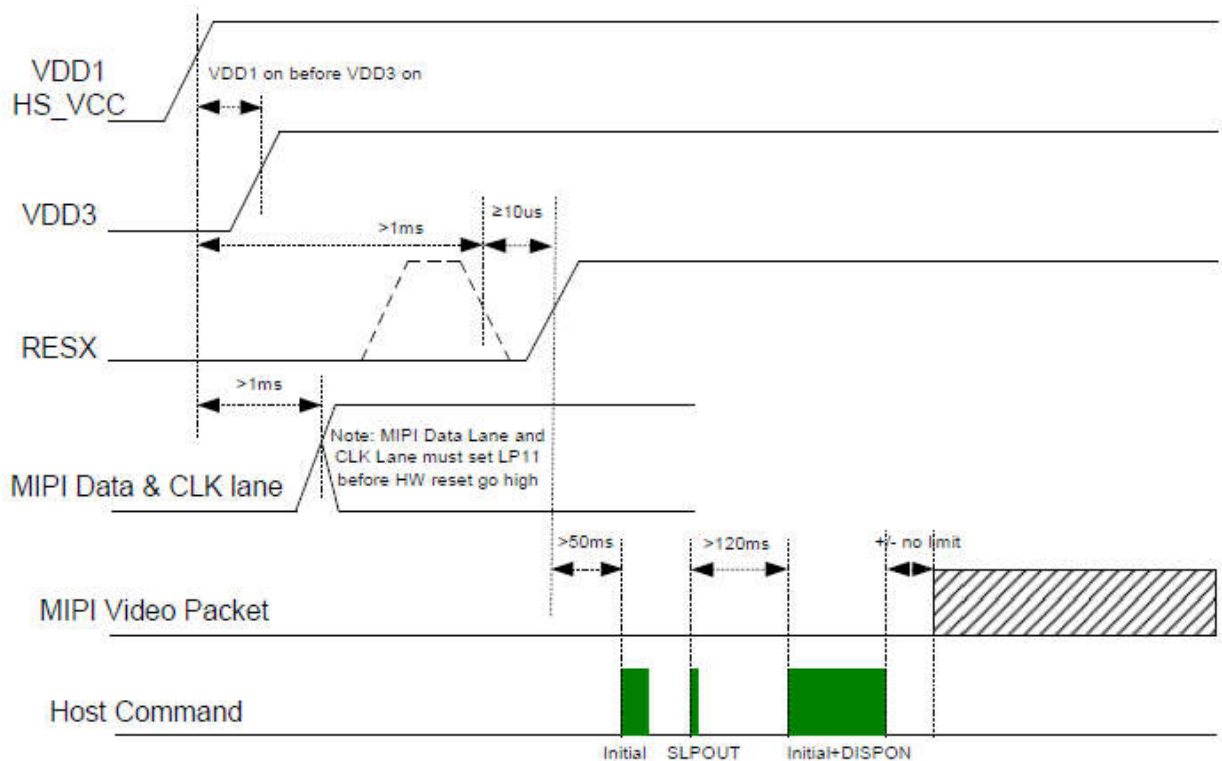
6.7 Power Sequence

The Power supply On/Off, Sleep In/Out and Display On/Off sequence is illustrated below.



Power On

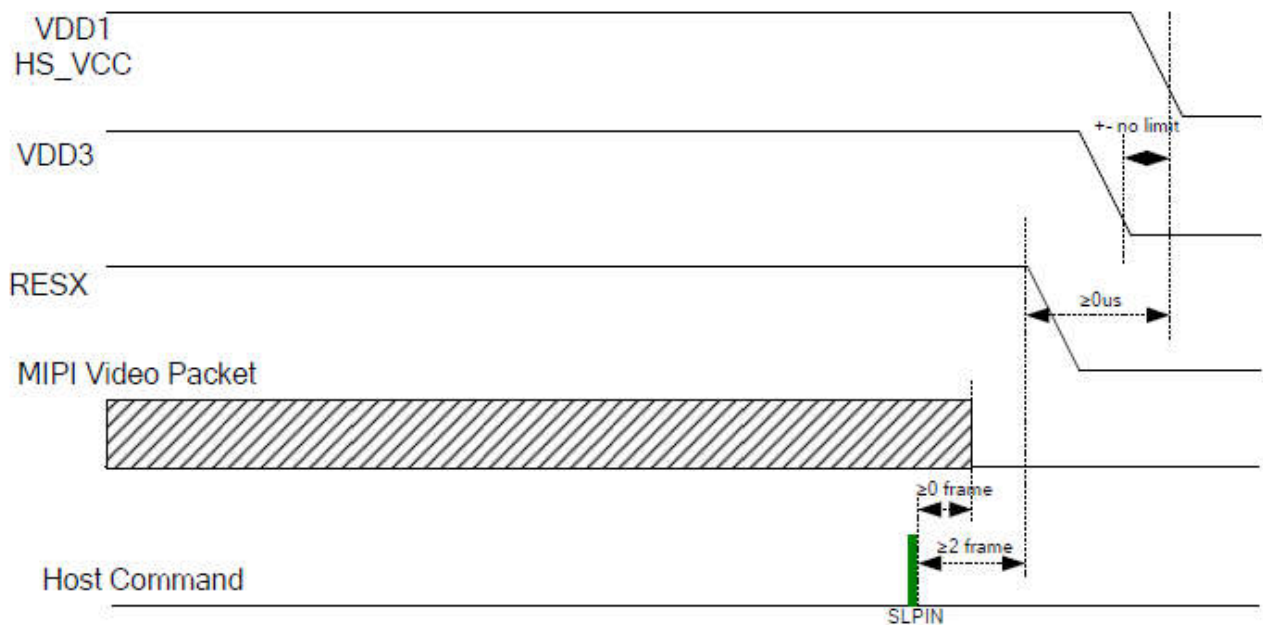
VDD1/VDD3 input power sequence (PCCS[1:0]="01", "10")



Note: SLPOUT CMD can before or after Initial code, but after SLPOUT CMD must wait at least 120ms

Power off

VDD1/VDD3 input power on sequence



6.8 Timing Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
MIPI data frequency	F_{DATA}	277	304	420	Mbps
Horizontal display area	T_{HD}	600			pixel
Hsync period time	T_H	675	741	901	pixel
Hsync pulse width	T_{HPW}	1	1	1	pixel
Hsync back porch	T_{HBP}	32	60	100	pixel
Hsync front porch	T_{HFP}	42	80	200	pixel
Vertical display area	T_{VD}	1024			H
Vsync period time	T_V	1085	1085	1225	H
Vsync pulse width	T_{VPW}	1	1	1	H
Vsync back porch	T_{VBP}	25			H
Vsync front porch	T_{VFP}	35	35	200	H

7. Optical specifications

Item	Symbol	Condition	Values			Unit	Remark
			Min.	Typ.	Max.		
Viewing angle (CR≥ 10)	θ_L	$\Phi=180^\circ$ (9 o'clock)	-	80	-	degree	Note 1
	θ_R	$\Phi=0^\circ$ (3 o'clock)	-	80	-		
	θ_T	$\Phi=90^\circ$ (12 o'clock)	-	80	-		
	θ_B	$\Phi=270^\circ$ (6 o'clock)	-	80	-		
Response time Rise+ Fall	T_{RT}	Normal $\theta=\Phi=0^\circ$	-	25	50	msec	Note 3
Contrast ratio	CR		600	800	-	-	Note 4
Color chromaticity	W_X		0.265	0.295	0.325	-	Note 2
	W_Y		0.285	0.315	0.345	-	Note 5 Note 6
Luminance	L		300	350	-	-	Note 6
Luminance uniformity	Y_U		75	80	-	%	Note 6,7
Frame Frequency	Frame	-	-	60	-	Hz	

Note 1: Definition of viewing angle range

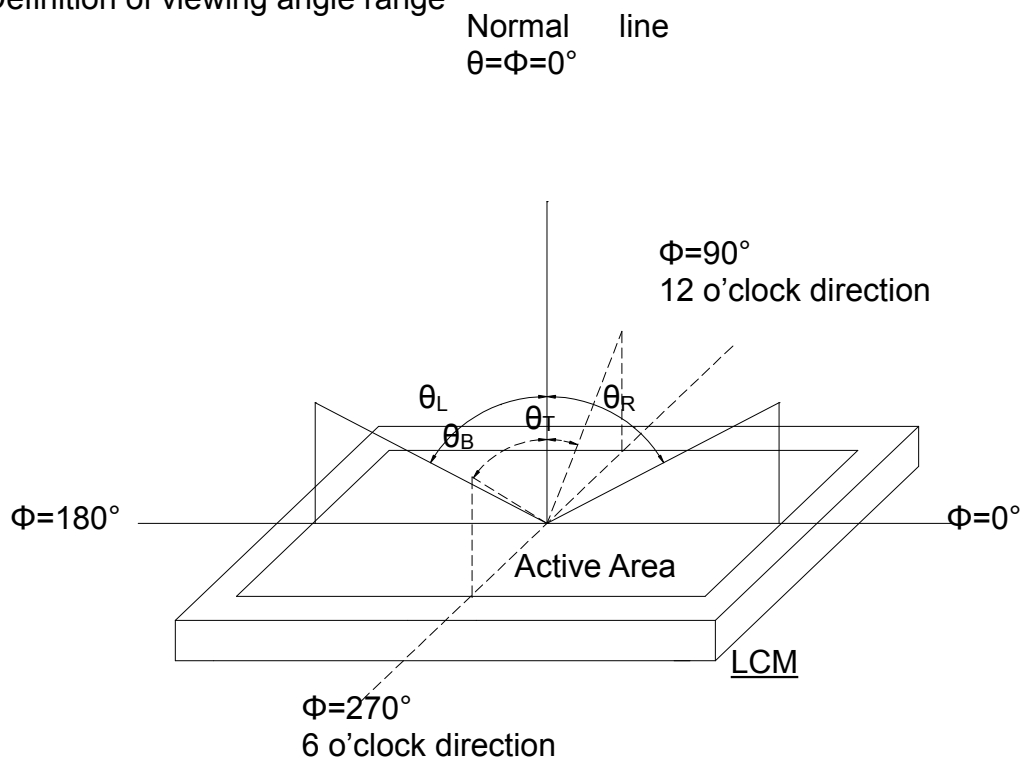


Fig. 4-1 Definition of viewing angle

Note 2: Definition of optical measurement system.

The optical characteristics should be measured in dark room. After 30 minutes operation, the optical properties are measured at the center point of the LCD screen. (Viewing angle is measured by ELDIM-EZ contrast/Height :1.2mm ,Response time is measured by Photo detector TOPCON BM-5A, other items are measured by BM-7A/Field of view: 1° /Height: 500mm.)

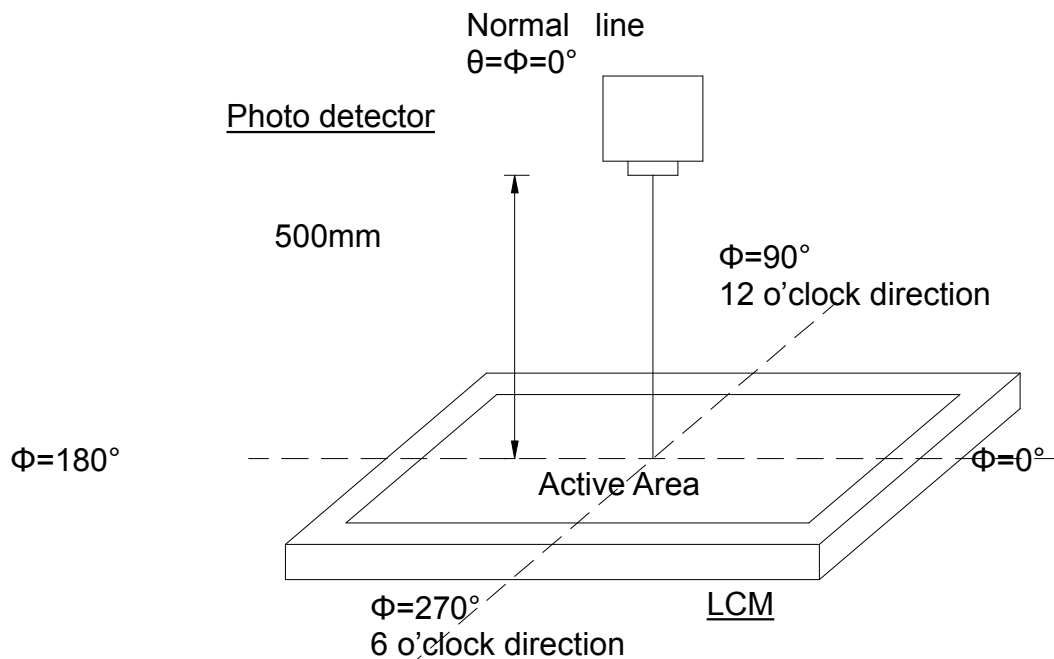


Fig. 4-2 Optical measurement system setup

Note 3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (T_{ON}) is the time between photo detector output intensity changed from 90% to 10%. And fall time (T_{OFF}) is the time between photo detector output intensity changed from 10% to 90%.

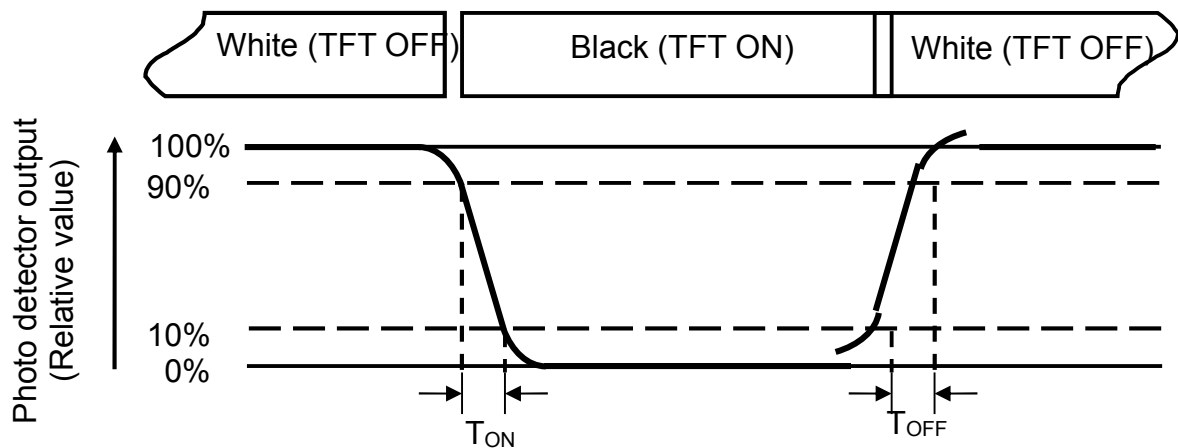


Fig. 4-3 Definition of response time

Note 4: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD on the "White" state}}{\text{Luminance measured when LCD on the "Black" state}}$$

Note 5: Definition of color chromaticity (CIE1931)

Color coordinates measured at center point of LCD.

Note 6: All input terminals LCD panel must be ground while measuring the center area of the panel. The LED driving condition is $I_{LED}=60\text{mA}$.

Note 7: Definition of Luminance Uniformity

Active area is divided into 13 measuring areas (Refer to Fig. 4-4).Every measuring point is placed at the center of each measuring area.

$$\text{Luminance Uniformity (Yu)} = \frac{B_{min}}{B_{max}}$$

L-----Active area length

W----- Active area width

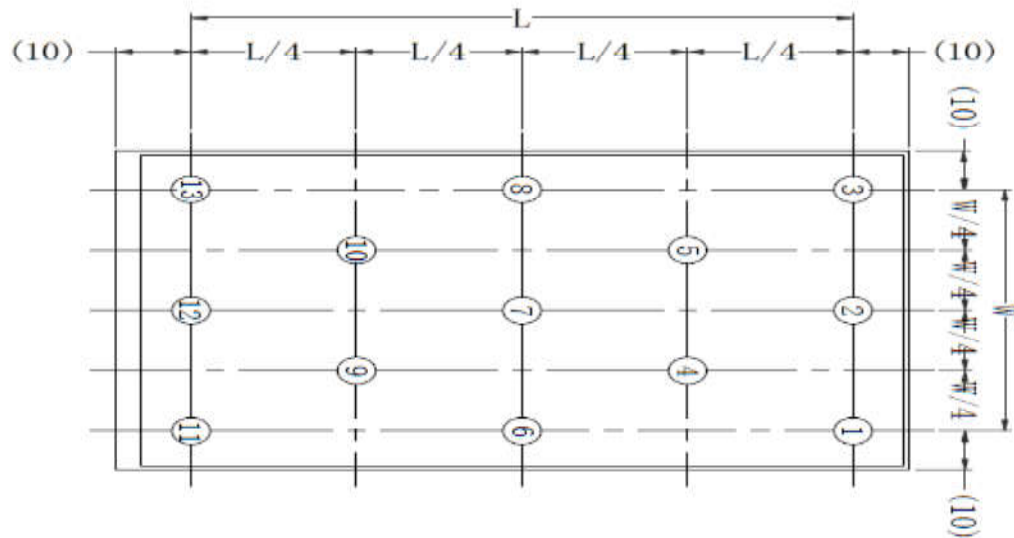


Fig. 4-4 Definition of measuring points

B_{max} : The measured maximum luminance of all measurement position.

B_{min} : The measured minimum luminance of all measurement position.

8. Reliability Test Conditions And Methods

Item	Test Conditions		Remark
High Temperature Storage	$T_a = 70^{\circ}\text{C}$	240 hrs	
Low Temperature Storage	$T_a = -30^{\circ}\text{C}$	240hrs	
High Temperature Operation	$T_s = 60^{\circ}\text{C}$	240hrs	
Low Temperature Operation	$T_a = -20^{\circ}\text{C}$	240hrs	
Operate at High Temperature and Humidity	60°C , 90%RH max.	240hrs	Operation
Thermal Shock	$-30^{\circ}\text{C} \sim +70^{\circ}\text{C}$ 100cycles 1Hrs/cycle		Non-operation
Electrostatic Discharge	Contact= $\pm 4\text{KV}$, class B Air= $\pm 8\text{KV}$, class B		

9. Handling Precautions

9.1 Mounting method

The LCD panel of K&D LCD module consists of two thin glass plates with polarizes which easily be damaged. And since the module in so constructed as to be fixed by utilizing fitting holes in the printed circuit board.

Extreme care should be needed when handling the LCD modules.

9.2 Caution of LCD handling and cleaning

When cleaning the display surface, Use soft cloth with solvent [recommended below] and wipe lightly

- Isopropyl alcohol
- Ethyl alcohol

Do not wipe the display surface with dry or hard materials that will damage the polarizer surface.

Do not use the following solvent:

- Water
- Aromatics

Do not wipe ITO pad area with the dry or hard materials that will damage the ITO patterns

Do not use the following solvent on the pad or prevent it from being contaminated:

- Soldering flux
- Chlorine (Cl) , Sulfur (S)

If goods were sent without being silicon coated on the pad, ITO patterns could be damaged due to the corrosion as time goes on.

If ITO corrosion happen by miss-handling or using some materials such as Chlorine (Cl), Sulfur (S) from customer, Responsibility is on customer.

9.3 Caution against static charge

The LCD module use C-MOS LSI drivers, so we recommended that you:

Connect any unused input terminal to Vdd or Vss, do not input any signals before power is turned on, and ground your body, work/assembly areas, assembly equipment to protect against static electricity.

9.4 packing

- Module employ LCD elements and must be treated as such.
- Avoid intense shock and falls from a height.
- To prevent modules from degradation, do not operate or store them exposed direct to sunshine or high temperature/humidity

9.5 Caution for operation

- It is an indispensable condition to drive LCD's within the specified voltage limit since the higher voltage then the limit cause the shorter LCD life.
- An electrochemical reaction due to direct current causes LCD's undesirable deterioration, so that the use of direct current drive should be avoided.
- Response time will be extremely delayed at lower temperature then the operating temperature range and on the other hand at higher temperature LCD's how dark color in them. However those phenomena do not mean malfunction or out of order with LCD's, which will come back in the specified operation temperature.
- If the display area is pushed hard during operation, some font will be abnormally displayed but it resumes normal condition after turning off once.
- A slight dew depositing on terminals is a cause for electro-chemical reaction resulting in terminal open circuit.

Usage under the maximum operating temperature, 50%Rh or less is required.

9.6 storage

In the case of storing for a long period of time for instance, for years for the purpose or replacement use, the following ways are recommended.

- Storage in a polyethylene bag with the opening sealed so as not to enter fresh air outside in it . And with no desiccant.
- Placing in a dark place where neither exposure to direct sunlight nor light's keeping the storage temperature range.
- Storing with no touch on polarizer surface by the anything else.
[It is recommended to store them as they have been contained in the inner container at the time of delivery from us

9.7 Safety

- It is recommendable to crash damaged or unnecessary LCD's into pieces and wash off liquid crystal by either of solvents such as acetone and ethanol, which should be burned up later.
- When any liquid leaked out of a damaged glass cell comes in contact with your hands, please wash it off well with soap and water

10. Precaution for use

10.1

A limit sample should be provided by the both parties on an occasion when the both parties agreed its necessity. Judgment by a limit sample shall take effect after the limit sample has been established and confirmed by the both parties.

10.2

On the following occasions, the handing of problem should be decided through discussion and agreement between responsible of the both parties.

- When a question is arisen in this specification
- When a new problem is arisen which is not specified in this specifications
- When an inspection specifications change or operating condition change in customer is reported to K&D , and some problem is arisen in this specification due to the change
- When a new problem is arisen at the customer's operating set for sample evaluation in the customer site.

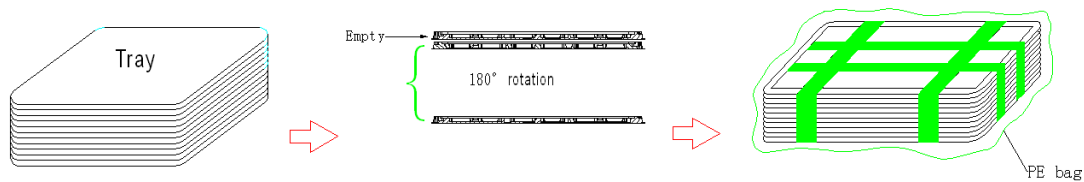
11. Package Drawing

LCM Product(Tray Type) Packing Flow Diagram

1.0 Packing BOM:

Please Reference the LCM BOM

2.0 Packing Procedure



First step

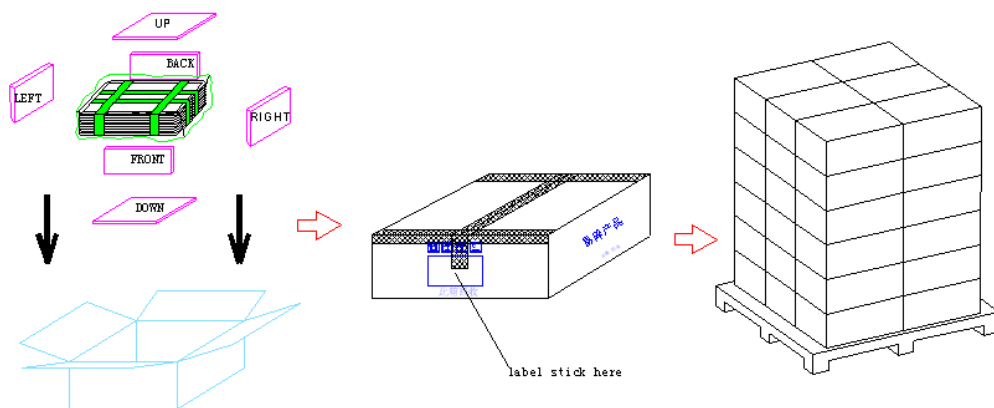
Putting products into the tray,
LCD A.A faces Upward,
(pay attention to the
direction)

Second step

Neighbouring trays should be
staggered 180° while stacking
up.
on the top,there is an empty tray
without product
the quality of layers please refer
to the BOM

Third step

Taping up the tray after
inspection,and put
them into a PE bag



Fourth step

Putting EPE foams and
products with trays into the
carton;
Close the carton box

fifth step

Sealing the carton with
cellulose tape ;
Stick on a carton label,

sixth step

Placing the boxes together
on a pallet (6 layers at most),

[illegible]

13. Bar code label and packing chest label

TBD

14. HSF Requirements

☒ RoHS (Restriction of the use of certain Hazardous Substances)

☐ HF (Halogen Free)

☐ REACH (Regulation concerning the Registration, Evaluation, Authorization and Restriction of Chemicals)

☐ Other regulations